

Logistic Regression

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Outline

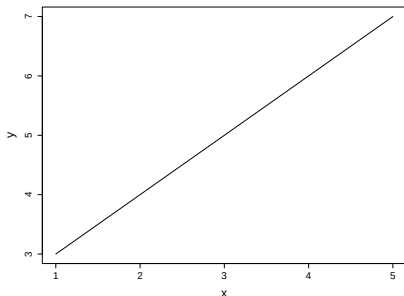
- 1 Intro to Logistic Regression
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Intro to Logistic Regression

Linear versus Logistic Regression

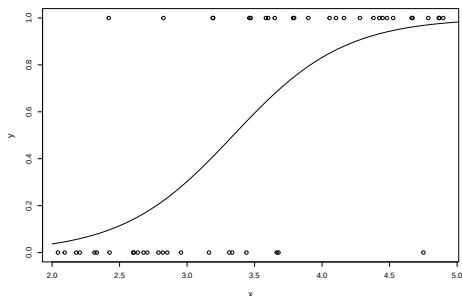
Linear Regression

- Relationship between continuous dependent variable and independent variables (continuous or categorical).
- Relationship of interest is linear.



Logistic Regression

- Relationship between **binary** dependent variable and independent variables (continuous or categorical).
- Relationship of interest (between probability of event occurring and independent variables) is no longer linear but rather logistic.



The logit transformation

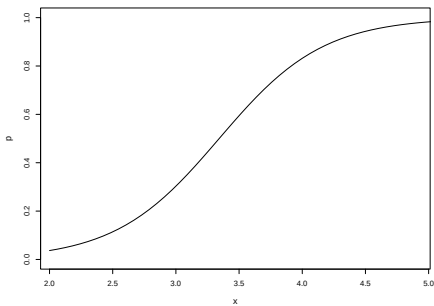
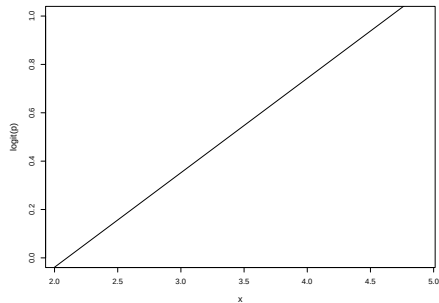
If we denote the i^{th} probability of the event of interest occurring as p_i , then the relationship between p_i and the independent variable is logistic (not linear). However we can apply the following transformation to p_i :

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right)$$

The logit transformed p_i now has a linear relationship with the independent variables as follows:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi}$$

$\text{logit}(p_i)$ is the log-odds of the event of interest occurring, and $\frac{p_i}{1-p_i}$ is the odds of the event occurring (ie. the probability of the event occurring over the probability of it not occurring).

p_i vs x  $\text{logit}(p_i)$ vs x 

Back-transforming $\text{logit}(p_i)$

It is important to remember that we do this transformation so that we can still fit a linear model to a non-linear relationship. BUT we are not actually interested in the log-odds of an event occurring, we are interested in the probability of it occurring.

So to get probabilities back out after we have fit our model we can back-transform the log-odds using:

$$p_i = \frac{e^{\beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi}}}{1 + e^{\beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi}}}$$

$$p_i = \frac{e^{\hat{LO}}}{1 + e^{\hat{LO}}}$$

We can simplify this to:

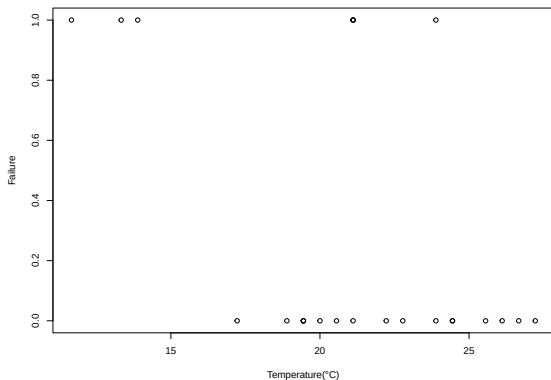
$$p_i = \frac{1}{1 + e^{-\hat{LO}}}$$

Example Problem

O-rings example

On 28 January 1986, the Space Shuttle Challenger exploded 73 seconds after lift-off due to O-ring (circular gasket that sealed the right rocket booster) failure caused by exposure to very low temperature (-0.5°C). To avoid this situation it is important to understand the relationship between temperature and the probability of O-ring failure.

	Temperature($^{\circ}\text{C}$)	Failure
1	11.67	1
2	13.33	1
3	13.89	1
4	17.22	0
5	18.89	0
6	19.44	0
7	19.44	0
8	19.44	0
9	20.00	0
10	20.56	0
11	21.11	0
12	21.11	1
13	21.11	1
14	21.11	1
15	22.22	0
16	22.78	0
17	23.89	0
18	23.89	1
19	24.44	0
20	24.44	0
21	25.56	0
22	26.11	0
23	26.67	0
24	27.22	0



Logistic Regression Analysis with Example

Performing Logistic Regression in R

```
fit <- glm(Failure-Temperature, data = orings, family = "binomial")
summary(fit)

##
## Call:
## glm(formula = Failure ~ Temperature, family = "binomial", data = orings)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.2125  -0.8253  -0.4706   0.5907   2.0512
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   5.3931     3.0535   1.766   0.0774 .
## Temperature  -0.3084     0.1502  -2.053   0.0400 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 28.975  on 23  degrees of freedom
## Residual deviance: 23.030  on 22  degrees of freedom
## AIC: 27.03
##
## Number of Fisher Scoring iterations: 4
```

- $\hat{\beta}_0 \longrightarrow$ Intercept estimate (on log-odds scale)
- $\hat{\beta}_1 \longrightarrow$ Temperature estimate (on log-odds scale)

So the regression equation for the example is given by:

$$\log\left(\frac{\widehat{p_i}}{1 - \widehat{p_i}}\right) = \hat{\beta}_0 + \hat{\beta}_1(\text{Temperature})$$

$$\log\left(\frac{\widehat{p_i}}{1 - \widehat{p_i}}\right) = 5.3931 - 0.3084(\text{Temperature})$$

Interpreting β coefficients

It is important to note the following change in the regression equation when transforming from log-odds to odds:

$$\log \left(\widehat{\frac{p_i}{1 - p_i}} \right) = \hat{\beta}_0 + \hat{\beta}_1(\text{Temperature}) \quad \widehat{\frac{p_i}{1 - p_i}} = e^{\hat{\beta}_0} \times e^{\hat{\beta}_1(\text{Temperature})}$$

In terms of log-odds On average, the log-odds of O-rings failure occurring decreases by 0.3084 units for each 1°C increase in temperature.

In terms of odds On average, the odds of O-rings failure changes by a factor of $e^{-0.3084} = 0.73$ for each 1°C increase in temperature, which translates to a 27% $((1 - 0.73) \times 100)$ decrease in the odds of O-ring failure occurring.

General interpretation on odds scale

- $\hat{\beta}_1 > 0 \longrightarrow e^{\hat{\beta}_1} > 1$

For example, let $\beta_1 = 0.8$

On average, the odds of the event occurring changes by a factor of $e^{0.8} = 2.226$ for each 1 unit increase in x , which translates to a 122.55% ($2.226-1$) increase in the odds the event occurring.

- $\hat{\beta}_1 < 0 \longrightarrow e^{\hat{\beta}_1} < 1$

For example, let $\beta_1 = -0.6$

On average, the odds of the event occurring changes by a factor of $e^{-0.6} = 0.549$ for each 1 unit increase in x , which translates to a 45.11% ($1-0.549$) decrease in the odds the event occurring.

- $\hat{\beta}_1 = 0 \longrightarrow e^{\hat{\beta}_1} = 1$

For this scenario there is no effect.

Odds-Ratios

Consider the case where our independent variable (x) is a dichotomous variable, where 1 = exposure and 0 = non-exposure. Eg. 1 = exposure to an advert and 0 = not exposed to the advert.

$$\log \left(\widehat{\frac{p_i}{1 - p_i}} \right) = \hat{\beta}_0 + \hat{\beta}_1 x_{1i}$$

The odds of the event among the unexposed are: $\frac{\widehat{p_{i0}}}{1 - \widehat{p_{i0}}} = e^{\hat{\beta}_0}$

The odds of the event among the exposed are: $\frac{\widehat{p_{i1}}}{1 - \widehat{p_{i1}}} = e^{\hat{\beta}_0 + \hat{\beta}_1}$

The ratio of the odds of the event for an exposed person relative to an unexposed person (the odds ratio) is:

$$OR = \frac{\frac{\widehat{p_{i1}}}{1 - \widehat{p_{i1}}}}{\frac{\widehat{p_{i0}}}{1 - \widehat{p_{i0}}}} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1}}{e^{\hat{\beta}_0}} = e^{\hat{\beta}_1}$$

For continuous variables the odds ratio represents the ratio of the odds of the event for an observation $X = x + 1$ relative to an observation $X = x$.

$$OR = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1(x+1)}}{e^{\hat{\beta}_0 + \hat{\beta}_1(x)}} = e^{\hat{\beta}_1}$$

So for a continuous variable the odds ratio is just the change in odds associated with a 1 unit increase in x .

Testing significance of estimated β coefficients

Step 1: Define the null hypothesis $H_0: \beta_1 = 0$

Step 2: Define the alternative hypothesis $H_a: \beta_1 \neq 0$

Step 3: Define the significance level $\alpha = 0.05$

Step 4: Calculate the test statistic

$$zscore = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} \sim N(0, 1) \quad zscore = \frac{-0.3084}{0.1502}$$

$$zscore = -2.053 \sim N(0, 1)$$

Step 5: Find the p-value

We can get the approximate p-value from the table: $p\text{-value} = 2 \times (0.5 - 0.4798) = 0.0404$

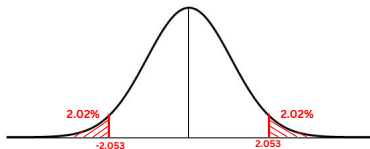


TABLE 1. STANDARD NORMAL DISTRIBUTION: Areas under the standard normal curve between 0 and z , i.e. $\Pr[0 < Z < z]$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857

```
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summary(fit)
```

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## Deviance Residuals:
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Confidence Intervals

We can get confidence intervals for our β estimates using

$$CI = \hat{\beta}_j \pm Z_{\frac{\alpha}{2}} \times se(\hat{\beta}_j)$$

We can calculate the 95% confidence interval for β_1 ($\beta_{\text{Temperature}}$) as follows:

$$CI = \hat{\beta}_1 \pm Z_{\frac{\alpha}{2}} \times se(\hat{\beta}_1)$$

$$CI = -0.3084 \pm 1.96 \times 0.1502$$

$$CI = [-0.603, -0.014]$$

```
confint(fit)
```

```
## Waiting for profiling to be done...
```

```
##                2.5 %      97.5 %
## (Intercept)  0.1769933 12.93198862
## Temperature -0.6802932 -0.05501466
```

```
confint.default(fit)
```

```
##                2.5 %      97.5 %
## (Intercept) -0.5916866 11.37787282
## Temperature -0.6027431 -0.01401074
```

Prediction

Predictions from a logistic regression model can be in any form (log-odds, odds or probabilities). However, we are usually interested in the predicted probabilities.

We can get these by first calculating the predicted log-odds:

$$\log \left(\widehat{\frac{p_i}{1-p_i}} \right) = \hat{\beta}_0 + \hat{\beta}_1(\text{Temperature})$$

$$\log \left(\widehat{\frac{p_i}{1-p_i}} \right) = 5.3931 - 0.3084(20) = -0.7749$$

And then back-transforming to get predicted probabilities:

$$p_i = \frac{1}{1 + e^{-\widehat{LO}}}$$

$$p_i = \frac{1}{1 + e^{-(-0.7749)}} = 0.315$$

We can do the same thing in R:

```
newdat <- list(Temperature=20)

# prediction
pred_logodds <- predict(fit,newdata = newdat, type = "link")
pred_logodds

##          1
## -0.7744449

pred_prob <- predict(fit,newdata = newdat, type = "response")
pred_prob

##          1
## 0.3155184
```

Using predictions for classification

When predicting with a logistic regression model, we are often interested in getting a classification for an individual (experiences event or not (1 or 0)).

To do this we can choose a threshold probability (π) and we assign all observations less than this threshold a 0 (do not experience event) and all observations above this threshold a 1 (does experience the event).

Confusion matrix

To assess the accuracy of our predictions, we can visualise the results in a confusion matrix.

		Observed		
		1	0	Total
Predicted	1	True Positive (tp)	False Positive (fp)	tp+fp
	0	False Negative (fn)	True Negative (tn)	fn+tn
Total		tp+fn	fp+tn	tp+fp+fn+tn

Positive Predictive Value

- Probability of the observation being positive given that the prediction is positive.
- $PPV = \frac{tp}{tp+fp}$

Negative Predictive Value

- Probability of the observation being negative given that the prediction is negative.
- $NPV = \frac{tn}{fn+tn}$

Sensitivity

- Ability of the model to predict a true positive.
- $Sensitivity = \frac{tp}{tp+fn}$

Specificity

- Ability of the model to predict a true negative.
- $Specificity = \frac{tn}{fp+tn}$

Sensitivity and Specificity trade-off

$$\pi = 0.2$$

		Observed		
		1	0	Total
Predicted	1	6 (tp)	8 (fp)	tp+fp
	0	1 (fn)	9 (tn)	fn+tn
Total		tp+fn	fp+tn	tp+fp+fn+tn

$$\text{Sensitivity} = \frac{tp}{tp+fn} = \frac{6}{6+1} = 0.857$$

$$\text{Specificity} = \frac{tn}{fp+tn} = \frac{9}{8+9} = 0.529$$

$$\pi = 0.8$$

		Observed		
		1	0	Total
Predicted	1	1 (tp)	0 (fp)	tp+fp
	0	6 (fn)	17 (tn)	fn+tn
Total		tp+fn	fp+tn	tp+fp+fn+tn

$$\text{Sensitivity} = \frac{tp}{tp+fn} = \frac{1}{1+6} = 0.143$$

$$\text{Specificity} = \frac{tn}{fp+tn} = \frac{17}{0+17} = 1$$

Confusion matrix in R

```
# get predicted probabilities
probs <- predict(fit, type = "response")

# create vector of zeros to store classification in
preds = rep(0, length(probs))
# if the probability is > 0.5 then replace the 0 in preds with a 1
#(can use a different threshold to 0.5)
preds[probs > 0.5] = 1
# preds now has our predicted classifications in it

# make confusion matrix
table(preds, as.numeric(orings$Failure))
```

```
##
## preds  0  1
##      0 16  4
##      1  1  3
```

Note that R by default orders the rows and columns with 0 first (no event) and not 1 (event).

Standard notation:

		Observed		
		1	0	Total
Predicted	1	True Positive (tp)	False Positive (fp)	tp+fp
	0	False Negative (fn)	True Negative (tn)	fn+tn
Total		tp+fn	fp+tn	tp+fp+fn+tn

R notation:

		Observed		
		0	1	Total
Predicted	0	True Negative (tn)	False Negative (fn)	fn+tn
	1	False Positive (fp)	True Positive (tp)	tp+fp
Total		fp+tn	tp+fn	tp+fp+fn+tn